Advanced Malware and Network Anomaly Detection

Today's goal • 3/3 items

Module 1
Course Introduction

- Course Overview
  Reading • 5 min
- Instructor Biography - Lanier Watkins
  Reading • 2 min

Module 2
Malware Threats Detection Part 1

Introduction to Malware Threat Detection Using AI

- Malware Threat Detection with AI
  Video • 4 min
- Reading References
  Reading • 15 min
- Introduction to Malware Threat Detection Using AI
  Practice Assignment • Grade: 100%

Basic Malware Analysis for Windows Operating Systems

- Malware Analysis for Windows OS
  Video • 3 min
- Reading References
  Reading • 15 min
- Basic Malware Analysis for Windows Operating Systems
  Practice Assignment • Grade: 100%

Module-end Assessments

- Self-Reflective Reading: Analyzing PE Files-Insights and Reflections
  Reading • 40 min
- Malware Threats Detection Part 1
  Graded Assignment • Grade: 98%

Module 3
Malware Threats Detection Part 2

Advanced Malware Detection with Clustering and Decision Trees

- Malware Threat Detection with Clustering and Decision Trees
  Video • 6 min
- Reading References
  Reading • 15 min
- Advanced Malware Detection with Clustering and Decision Trees
  Practice Assignment • Grade: 92%

Techniques for Metamorphic Malware Detection

- Metamorphic Malware Detection
  Video • 10 min
- Reading References
  Reading • 15 min
- Techniques for Metamorphic Malware Detection
  Practice Assignment • Grade: 96%

Module-end Assessments

- Self-Reflective Reading: Deep Dive into AI and Malware Detection
  Reading • 40 min
- Practice Lab: SMS Spam Collection Using Naïve Bayes Spam Filter
  Lab • 1h
- Malware Threats Detection Part 2
  Graded Assignment • Grade: 100%

Module 4
Network Anomaly Detection with AI

Introduction to Network Anomaly Detection

- Network Anomaly Detection
  Video • 5 min
- **Reading References**
  Reading • 30 min
- Introduction to Network Anomaly Detection
  Practice Assignment • 15 min

Anomaly Detection with Botnet Data: A Case Study

- Anomaly Detection: Botnet Case Study
  Video • 9 min
- Reading References
  Reading • 30 min
- Optional Reading
  Reading • 30 min
- Anomaly Detection with Botnet Data: A Case Study
  Practice Assignment • 15 min

# Reading References

0:13 / 0:33 1.5x

This week's reading assignment includes the following:

- **Textbook:** Parisi, A. A. (2019). Hands-On Artificial Intelligence for Cybersecurity. Packt Publishing.
  - Chapter 5
- **Conference Paper:** Manoj, V., Wenda, S. Sihan, N., Rouff, C., Watkins, L., & Rubin, A. (2023). Explainable Autonomic Cybersecurity for Industrial Control Systems. *2023 IEEE 13th Annual Computing and Communication Workshop and Conference (CCWC)*, pp. 900-906. https://ieeexplore.ieee.org/document/10099307

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